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Fig. 1. Photographs of the LEGO landscape build representing *Filoboletus manipularis*. The top photograph shows the model under normal lighting conditions, while the bottom photograph shows the illuminated effect in darkness.

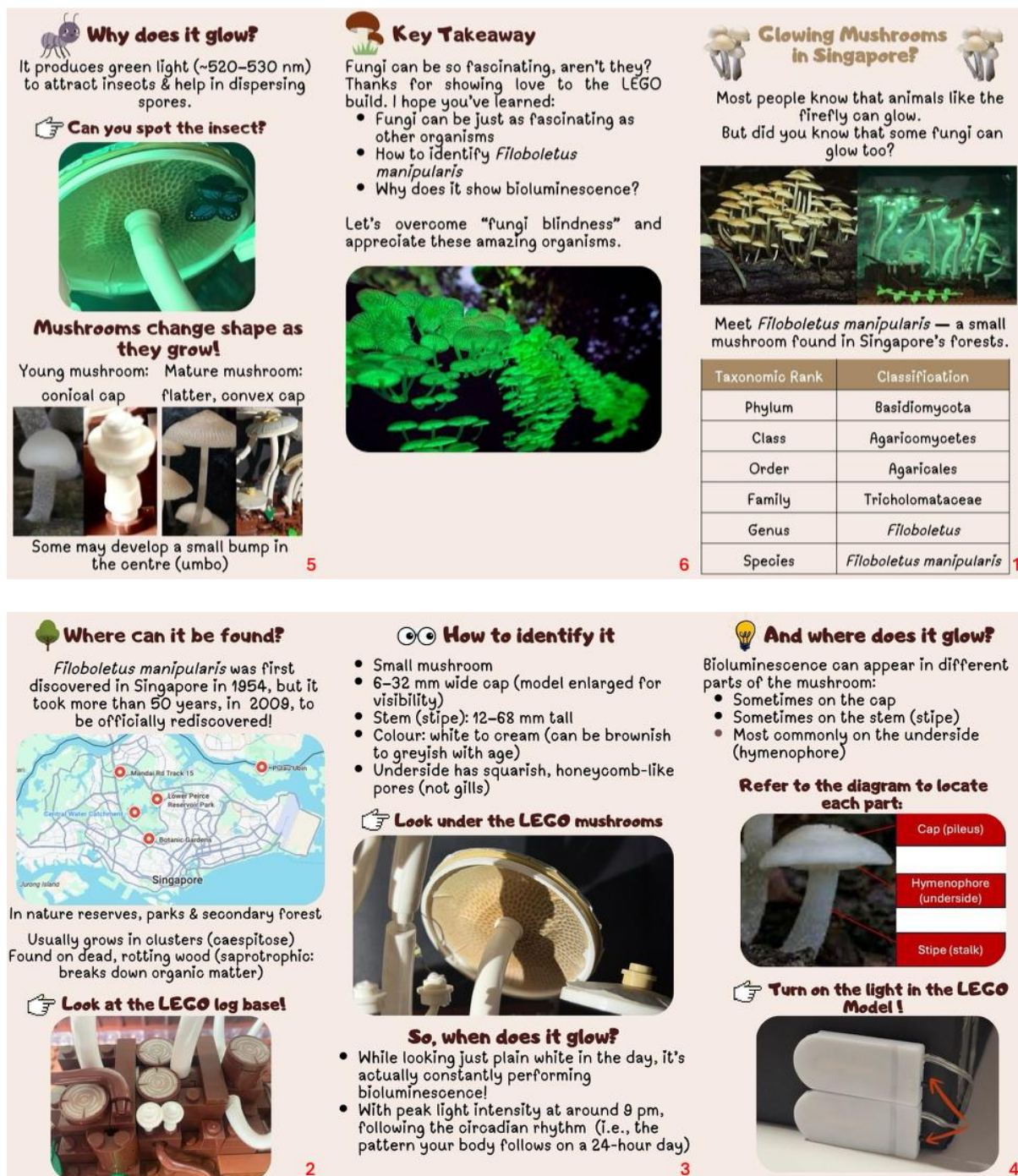


Fig. 2. Photograph showing the general outline of the brochure provided alongside the LEGO build to guide and educate audiences on *Filoboletus manipularis*. Page numbers are indicated in bright red in the right-hand corner.

This assignment aims to explore and educate others about the fascinating phenomenon of fungal bioluminescence through a physical LEGO based landscape model (Fig. 1). While many people are familiar with bioluminescence in animals, most notably fireflies, which emit light through a chemical reaction involving the enzyme luciferase, oxygen, and the substrate luciferin (Rodda, 2012), it is likely that far fewer are aware that certain fungi exhibit similar

properties. Using this as a basis, this project highlights the often-overlooked complexity and ecological significance of fungi.

The species featured in this model is *Filoboletus manipularis*, a member of the phylum Basidiomycota (The Biodiversity of Singapore, n.d.). This species is found throughout Singapore's nature reserves and parks and has been documented in several specific locations, including Lower Pierce Reservoir Park, Pulau Ubin, Singapore Botanic Gardens and the Central Catchment Nature Reserve (Lee, 2022; Lee & Choong, 2023). Although it was first discovered in Singapore in 1954, it was only examined and rediscovered again in 2009. This reflects the broader issue of "fungi blindness," where fungi are often underappreciated in comparison to other organisms, thereby supporting the need to integrate greater public awareness and attention towards this species.

The LEGO model is designed to represent both the natural habitat and growth behaviour of *Filoboletus manipularis*. Wooden printed tiles and a log-based structure are used to simulate decomposing wood, while multiple mushroom builds are arranged in clustered formations. This reflects how the species typically grows gregariously to caespitose on fallen wood in secondary forests (Lee, 2022; Lee & Choong, 2023), highlighting its saprotrophic role in breaking down organic matter.

To enhance audience engagement and learning, a brochure is placed alongside the model (Fig. 2) to guide viewers through the display. It provides additional biological information, including the dimensions of the fruiting bodies, which range from 12 to 68 mm in stipe height and 6 to 32 mm in pileus diameter (Lee & Choong, 2023). Due to constraints in available LEGO pieces and the wide natural variation of the species, the mushroom caps are scaled up in the model to better illustrate size differences. While this introduces some inaccuracy in scale, it improves visual accessibility and educational effectiveness.

In addition, this assignment aims to teach audiences how to identify this species in the wild. Identification can be achieved by examining the underside of the fruiting body, which features a distinctive network of squarish, honeycomb-like symmetrical white pores, with approximately 1 to 3 pores per millimetre (Lee, 2022; Leonard, 2021). This characteristic is printed and represented in the LEGO build and can be observed by viewing the underside of the constructed mushrooms. Moreover, emphasis is also placed on how the pileus shape may vary with age, appearing more conical in younger specimens and becoming flatter or more convex as the mushroom matures (Lee, 2022). This variation is illustrated in the LEGO model through the use of differently shaped mushroom caps to represent younger and more mature fruiting bodies.

Finally, the bioluminescent aspect of this model serves as a key educational component, focusing on where, when, and why this species emits light. The accompanying brochure will explain that bioluminescence in *Filoboletus manipularis* is not restricted to a single anatomical structure and may occur in the stipe, hymenophore, and sometimes even the pileus (Lee & Choong, 2023). However, greater luminosity has been observed in the hymenophore (Lee, 2022), which is why the lighting in the model is positioned beneath the cap.

The materials also explain when bioluminescence occurs. Although the species may appear plain white and unnoticeable during the day (Rodda, 2012), studies have shown that fungal bioluminescence is continuously produced and follows a circadian rhythm, with peak

intensity occurring at night around 9 pm (Perry et al., 2024). In addition, the choice of green lighting in the model reflects the actual emitted wavelength of approximately 520 to 530 nm, which lies within the green spectrum (Lee, 2022).

With regard to why this phenomenon occurs, while the specific function of bioluminescence in *Filoboletus manipularis* has not been directly studied, research on other bioluminescent fungi suggests that light emission may play a role in attracting phototactic insects that aid in spore dispersal (Ke & Tsai, 2021; Lu et al., 2024). This concept is illustrated in the model through the inclusion of a small insect figure positioned near the light source, symbolizing the interaction between fungi and insects in facilitating spore dispersal.

Through this interactive and visual representation, this assignment seeks to achieve three key educational objectives. First, to demonstrate that fungi can be as fascinating and complex as more commonly appreciated organisms like fireflies. Second, to provide knowledge on how to identify *Filoboletus manipularis* in the wild. And third, to explain the ecological and biological significance of fungal bioluminescence.

Ultimately, this creative assignment aims to challenge the common neglect of fungi in biological appreciation and hopes to encourage a deeper interest in these remarkable organisms.

References

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